

Final Exam: Database Management Systems – SQL Section

Part B – Intermediate SQL (JOIN, GROUP BY, Subquery – Solutions)

Schema:
Students(student_id, name, dob, department_id)
Departments(department_id, department_name)
Courses(course_id, course_name, department_id)
Enrollments(enrollment_id, student_id, course_id, grade)

6. Retrieve the name and department_name of all students:

```
SELECT s.name, d.department_name
FROM Students s
JOIN Departments d ON s.department_id = d.department_id;
```

7. Find the average grade of each course. Only show courses with an average grade greater than 3.0:

```
SELECT c.course_name, AVG(e.grade) AS avg_grade
FROM Enrollments e
JOIN Courses c ON e.course_id = c.course_id
GROUP BY c.course_name
HAVING AVG(e.grade) > 3.0;
```

8. Display all students who have not enrolled in any course:

```
SELECT s.student_id, s.name
FROM Students s
WHERE s.student_id NOT IN (
    SELECT e.student_id FROM Enrollments e
);
```

9. Display the course_name and the number of students enrolled in each course, ordered from highest to lowest:

```
SELECT c.course_name, COUNT(e.student_id) AS num_students
FROM Courses c
LEFT JOIN Enrollments e ON c.course_id = e.course_id
GROUP BY c.course_name
ORDER BY num_students DESC;
```

10. List all students whose grade in any course is above the average grade of that course:

```
SELECT s.student_id, s.name, c.course_name, e.grade
FROM Enrollments e
JOIN Students s ON e.student_id = s.student_id
JOIN Courses c ON e.course_id = c.course_id
WHERE e.grade > (
    SELECT AVG(e2.grade)
    FROM Enrollments e2
    WHERE e2.course_id = e.course_id
);
```

